

SplitUPI - The Complete Guide

One file. Web app + Android app + tests + robots. Simple words, nothing skipped.

pranvirsingh/split-upi | v1.0 to v1.4 | Sep 2026

Part A - Your setup

1. What you own (3 things)

- GitHub repo: github.com/pranvirsingh/split-upi. Home of all code. Public means anyone can read it, only you can change it.
- Vercel site: split-upi-ochre.vercel.app. The live website. Rebuilds itself on every merge to main.
- Releases: frozen milestones v1.0 to v1.4. v1.4 also carries `splitupi-v1.4.0.apk`, the Android installer.
- Rule: the repo is the truth. The site and the APK are built from it, never by hand.

2. Tools on your machine

- Node.js: runs JavaScript outside the browser. npm comes with it and installs code packages. Check: `node --version`.
- Git: save-points for code. Every commit is a point you can return to. Check: `git log --oneline -5`.
- gh: drives GitHub from your terminal (repos, PRs, releases, checks). Already logged in as you.
- Editor: any code editor for the "UPI App" folder. No Android Studio, no Mac needed for anything we do.

3. The only 6 commands you need

- `npm run dev` -> run the site on your machine to see and click it.
- `npm test` -> 39 automatic checks. `npm run test:logic` -> 16 spec checks.
- `npm run build` -> packs the site into `dist/` for publishing. Must pass before any release.
- `git status` -> what changed. `gh pr checks <n>` -> is the robot happy. `gh release view v1.4.0` -> what is published.

Part B - The web app

4. Folder map (what lives where)

- `src/lib/` = brain. `split.ts` cuts money. `upi.ts` builds payment links. `validation.ts` checks input. `format.ts` writes Rs. text. `download.ts` makes PNGs. `native.ts` is the app bridge.
- `src/components/` = face. Header, PaymentForm, AmountPreset, PaymentSummary, QRCodeCard, Disclaimer, Footer. Each shows what `App.tsx` tells it.
- `src/App.tsx` = boss. Holds all data, runs generate, arranges the screen. `src/main.tsx` + `index.html` = the door.
- `public/` = shipped as-is: `privacy.html`, icons, manifest, offline helper `sw.js`.
- `tests/` + `scripts/` = guards. `.github/workflows/` = robots. `.opencode/` = our work rules. `assets/` = icon sources.

- Never touch by hand: node_modules/ (installed), dist/ (built), android/app/build/ (built), android/.../assets/ (synced copy).

5. One full run, step by step

- Step 1: you type UPI id (mohit@upi), name, total (4500), max-per-QR (1999). Typing only saves text.
- Step 2: press generate. The checker asks: one @ in id? name present? amounts positive, max 2 decimals? parts within 50 QRs?
- Step 3: bad input -> friendly red message, no QRs. Good input -> parts list [1999, 1999, 502].
- Step 4: each part becomes a link:
upi://pay?pa=mohit@upi&pn=Mohit%20Kumar&am=1999.00&cu=INR&tn=Invoice%20001.
- Step 5: each link becomes a QR image. Scan with any UPI app -> payment screen opens, amount filled.
- Step 6: buttons. Download PNG (pretty branded card), Share, Copy link, Print all, Download all, Share all.

6. The brain files

- split.ts: math in paise (integers) so 0.1 + 0.2 style bugs are impossible. sumAmounts() double-checks totals.
- upi.ts: formatUpiAmount() forces 2 decimals (502 -> 502.00). buildUpiUri() encodes spaces/symbols, skips empty note, keeps @ readable.
- validation.ts: isValidUpid() + validateInputs(). Returns ok + parts, or ok=false + exact error per field.
- format.ts: Indian grouping (100000 -> 1,00,000). download.ts: qrDataUrl(), branded canvas PNG, copy/share helpers.
- constants.ts: MAX_TOTAL_AMOUNT = 10,00,000. MAX_QR_COUNT = 50. Presets 499/999/1999/4999 = convenience only, never official limits.
- DISCLAIMER_TEXT: legal lines. Never delete, never promise avoiding fees or limits.

7. The face files

- Idea: App.tsx is the parent and holds all data. Components are children: they show data and report clicks back up.
- PaymentForm: inputs, red errors, preset pills, big button. PaymentSummary: totals line, bulk buttons, QR grid, empty state.
- QRCodeCard: one QR + amount + id + png/share/copy/open + tiny popup message. Header: top bar + status + generate button.
- Footer: status bar; disclaimer/privacy/faq hide in expanders so the page stays short.

8. Looks

- Tailwind: style words inside the HTML (bg-forest, text-mist). No per-page style files.
- Theme tokens (index.css): night #0c1210, panel #121a15, mist #ece7d9, forest #173b2e, gold #d3a62c.

- Fonts: Inter for text, JetBrains Mono for console labels. QR squares always sit on white so phones can scan.
- Print rules: screen buttons hide on paper; a hidden sheet with big QRs appears instead.

9. PWA, offline, SEO, privacy page

- manifest.webmanifest: name, icons, colors. Makes the site installable from the browser.
- sw.js: saves the app shell on first visit, so the generator works offline after that.
- index.html title + description: what Google shows. FAQ text on the page feeds search.
- public/privacy.html: "data stays on device" in plain words. Linked in the footer. Play Store will require this URL.

Part C - Tests and robots

10. Tests (safety net)

- tests/split.test.ts: 4500/1999, 10000/1999, 1999/1999, 1998/1999 + paise + sum-exact + zero/negative rejection.
- tests/upi.test.ts: exact spec URI, note skipped when empty, symbols encoded, spaces trimmed.
- tests/validation.test.ts: good form passes; 0, negatives, 3-decimals, bad ids, missing name, 100-QR overflow all refused with messages.
- tests/qrcode.test.ts: builds real QR PNGs, decodes them with jsQR, proves each carries its exact link. This is the star test.
- tests/format.test.ts: 4500 -> 4,500 and 1,00,000 grouping. scripts/test-logic.mjs: 16-check spec list, also runs in CI.
- To add a test: copy one it(...) block, change values, run npm test. Green in ~2 seconds.

11. Robots and rules

- CI (.github/workflows/ci.yml): every PR runs npm ci -> test:logic -> npm test -> build. Any red = merge blocked.
- CD = Vercel: merge to main rebuilds the site. Each PR gets its own preview link to click through.
- main is locked: green CI required, branch must be fresh, no direct pushes, no force-push, no delete.
- Skill splitupi-release: branch feat/su-*, short commits, human PR text, green checks, squash merge, tag vX.Y.Z, release notes.
- Agents: web-dev (site work), mobile-dev (app work), qa-tester (checks). /ship runs the whole track.

12. Version story (verify each yourself)

- v1.0: first live site. Check: release page + live URL loads.
- v1.1: forest-gold colors + README live link. Check: gold buttons, live link at README top.
- v1.2: console rebuild, half the words. Check: dark panes, \$ input/\$ output, status-bar footer.
- v1.3: 39 tests + QR-decode, zero app change. Check: tests/ folder + CI runs npm test.
- v1.4: Android app + privacy page. Check: splitupi-v1.4.0.apk on the release.

- Rule kept every time: one version = one job.

Part D - Android from zero

13. What an Android app is

- APK = install file for phones (like .exe for Windows). Ours: splitupi-v1.4.0.apk on the v1.4 release.
- AAB = the file Play Store wants later. Same app, different wrapper. Not needed yet.
- To install ours: download APK on phone -> tap -> allow "install unknown apps" once -> open SplitUPI icon.
- Inside every app: an Activity = one screen window. Ours: MainActivity.java (5 lines) showing our site full-screen.
- AndroidManifest.xml = the id card: app name SplitUPI, id app.splitupi, icon, splash screen, no extra permissions.
- res/ = looks: mipmap-* icons per screen size, drawable-* splash per screen size. All generated, never hand-drawn.

14. Java, SDK, Gradle (one line each, then ours)

- Java = the language Android is built in. Our Java is 5 lines; Capacitor ships the other 99%.
- SDK = Google parts box (platform android-36, build-tools 36). Lives on GitHub machines. You install nothing.
- Gradle = the builder robot. Reads *.gradle recipes, downloads pieces, outputs the APK. Command: ./gradlew assembleDebug.
- Ours: minSdk 24 = runs on Android 7 and newer (almost all phones). Java 21 required by Capacitor 8.
- Key files: android/build.gradle (project recipe), android/app/build.gradle (app recipe), android/variables.gradle (versions in one place).

15. Capacitor, the wrapper

- Picture a shop window: your website inside, Android frame outside. Users see an app; inside it is our site.
- capacitor.config.ts has 3 lines that matter: appName SplitUPI, appId app.splitupi, webDir dist.
- npx cap sync android: copies dist/ (fresh site) into android/app/.../assets. CI runs it before every APK.
- Plugins = extra powers. Filesystem (write files) + Share (system sheet). Installed via npm, registered automatically.

16. Our 3 native things

- Two faces: isNative() asks "app or website?" The app hides the marketing masthead + steps, keeps the generator, adds safe-area padding.
- Save/share: browsers can download files, apps cannot. So saveAndShareImage() writes the PNG to private cache (needs no permission) then opens the share sheet: Photos, WhatsApp, Drive.

- Look: assets/icon.png + splash.png (gold S on forest/night) became 123 sized files via capacitor-assets. Branding done once, forever.
- Read the whole bridge: src/lib/native.ts is 33 lines. Smallest important file in the project.

17. How the APK is born

- Push tag v1.x -> Android robot wakes on GitHub: Java 21 + SDK 36 -> npm ci -> build site -> cap sync -> gradle APK.
- APK uploads to the release by itself (needs contents:write permission, already set).
- Manual rerun anytime: gh workflow run Android --ref main -f tag=v1.4.0. No new version needed.
- Our 4 real failures and fixes: broken SDK action (use runner SDK), gradlew lost exec bit (chmod), Java 17 too old (use 21), upload 403 (grant permission).
- Play Store later needs: signed AAB (secret key file), store listing text + screenshots, this privacy URL. Nothing to do now.

Part E - Check yourself

18. Reading order (30 minutes)

- 1) src/lib/split.ts (40 lines, the math). 2) src/lib/upi.ts (30 lines, the links). 3) src/lib/validation.ts (the rules).
- 4) src/lib/native.ts (33 lines, the bridge). 5) src/App.tsx generate function (the flow). 6) capacitor.config.ts (the wrapper).
- 7) .github/workflows/android.yml (the APK robot). 8) README.md (the index of everything).

19. Words to know

- Repo / branch / PR / tag / release: home / safe copy / proposed change / frozen milestone / milestone + notes + files.
- CI / CD: test robot / publish robot. npm / Vite / React: installer+runner / site packer / face builder.
- UPI link / QR / paise math: payment text / text-as-squares / integer money without decimal bugs.
- Manifest / Activity / Gradle / SDK: app id-card / app window / app builder / Google parts box.
- APK / AAB / PWA: phone installer / store upload / installable website. Cache dir: private storage, no permission.
- Squash merge: many small commits become one clean history line. Preview link: try a PR before merging.

20. If lost, do this

- 1) Find its chapter above. 2) Open the file listed. 3) Run npm test (code?) or gh pr checks (pipeline?).

Done. This file is your single truth. When the app grows, this guide grows with it.